ANA FERNÁNDEZ BARANDA

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EDUCATION

PhD in Applied Mathematics *October 2022 - July 2026*
CMAP, École Polytechnique
Title: Stochastic modeling of hematopoiesis and Myeloproliferative Neoplasms: hydrodynamic limit and statistical analysis
Advisors: Sylvie Méléard and Vincent Bansaye

M.Sc. in Applied Mathematics (PhD Track) *September 2020 - September 2022*
Institut Polytechnique de Paris
Master scholarship from IP Paris
M1 Mathematics Jacques Hadamard
M2 Mathematics for living sciences

B.Sc. in Applied Mathematics *August 2015 - December 2019*
Instituto Tecnológico Autónomo de México

Scientific Baccalauréat *June 2015*
Lycée Franco-Mexicain

TEACHING EXPERIENCE

Discrete Mathematics *Spring 2023, 2024, 2025*
École Polytechnique, Palaiseau, France.
Tutorial classes for first year students of the Bachelor program.

Mathematics *August 2018 - May 2019*
ITAM Construye program, Mexico City, Mexico.
Volunteer teacher to help students of elementary and secondary school standardize their level in mathematics.

RESEARCH INTERESTS

- Stochastic Processes
- Statistics
- Mixed-Effects Models
- Mathematical Modelling
- Mathematical Biology

PUBLICATIONS

Preprints

[1] Bansaye, V., **Fernández Baranda, A.**, Giraudier, S., Méléard, S. *Hematopoiesis as a continuum: from stochastic compartmental model to hydrodynamic limit.* (2026). https://arxiv.org/abs/2601.19307

[2] **Fernández Baranda, A.**, Bansaye, V., Lauret, E., Mounier, M., Ugo, V., Méléard and Giraudier, S. *Classical JAK2V617F+ Myeloproliferative Neoplasms emergence and development based on real life incidence and mathematical modeling.* (2024). https://arxiv.org/abs/2406.06765

TALKS AND POSTERS

Talks

- **Junior Female Researchers in Probability**
  June 2026, Berlin, Germany.
- **Journées Math Bio Santé**
  November 2025, Montpellier, France.
- **Mathematical Biology Modelling days of Besançon**
  October 2025, Besançon, France.
- **École de printemps de la chaire Modélisation Mathématique et Biodiversité**
  June 2024, Aussois, France.

Posters

- **Journées Math Bio Santé**
  November 2023, Marne la Vallée, France.

PRE-PHD RESEARCH EXPERIENCE

M2 Research Internship *April 2022 - August 2022*
École Polytechnique
Title: Characterization of cancer detection time as a function of biological parameters
Supervisors: Sylvie Méléard and Vincent Bansaye

M2 Project *October 2021 - March 2022*
Title: A simple model for the characterization of biological networks
Supervisor: Gael Raoul

M1 Research Internship *April 2021 - July 2021*
Université de Lorraine
Title: Study of the SIR model with a lockdown type control
Supervisor: Édouard Strickler

M1 Project *January 2021 - March 2021*
Title: Invasion in random environments
Supervisor: Vincent Bansaye

Undergraduate Dissertation

Title: Some spatio-temporary properties in a model of the PU.1-GATA-1 interaction
Supervisor: Víctor Francisco Breña Medina

TECHNICAL SKILLS

Computer languages: Python, R, Julia, MATLAB, Java
Software and tools: Monolix

LANGUAGES

Spanish Native
French Advanced
English Advanced